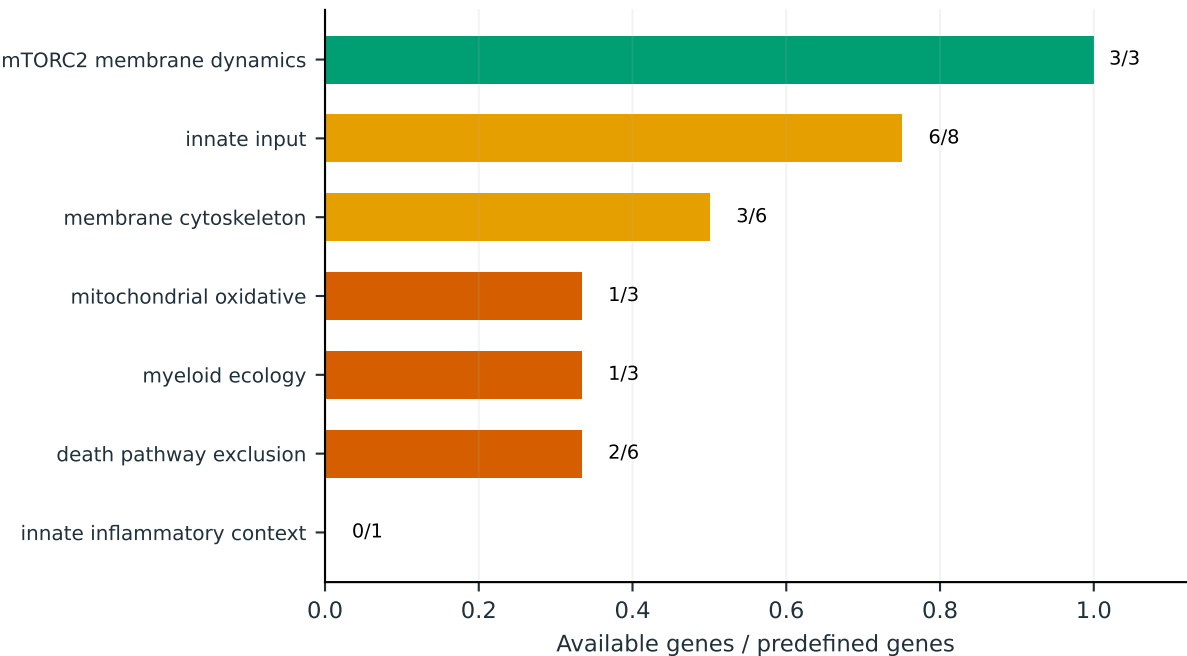
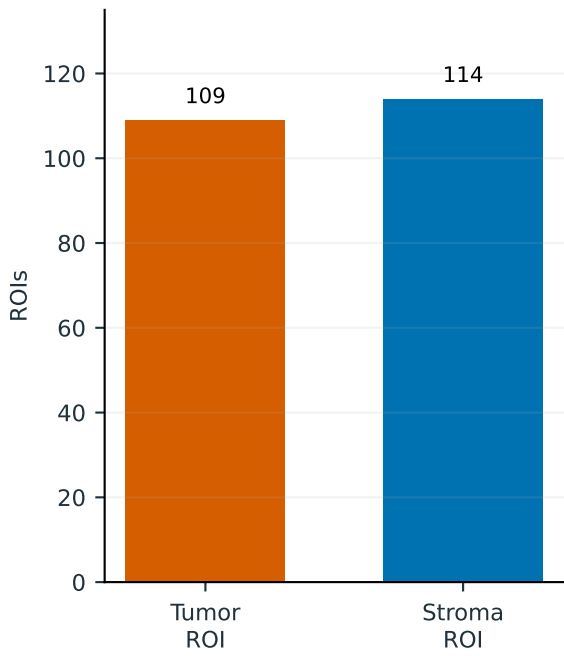


Supplementary Figure S4. GeoMx coverage audit and ROI-level spatial constraints

A GeoMx module coverage



B ROI composition
40 paired patients; 223 ROIs



C Interpretation boundaries

BAX-only mitochondrial proxy

Only BAX is available
(1/3 final Mito3 genes); this is not Mito3.

Myeloid proxy

Only SPP1 is available
(1/3 genes); it is not a complete myeloid-ecology score.

No geometric ROI coordinates or cell segmentation:
paired-ROI differences do not prove spatial adjacency or cell contact.